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Lost and Found: Identifying Objects in Long-term Surveillance Videos

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***Abstract*—What good are surveillance videos without knowing what objects are there? Object classification has been actively researched for images and more recently, for videos, but not in the long-term sense. Videos that span a long period of time has its arduous challenges in such a task. This paper intends to bridge that gap by exploring object classification in long-term surveillance videos. In this work, we introduce a complete framework for processing long-term surveillance videos with the aim of classifying moving objects into five distinct classes commonly found in these scenes. With effective extraction of moving objects and track creation, object features are then encoded in a bag-of-words model before performing classification. Extensive experiments were conducted on a selected portion of the recent LOST dataset. With state-of-the-art PHOW features, we are able to achieve the highest accuracy of around 92% using a track-based classification scheme that is robust against potential frame-level misclassifications.**

I. Introduction

Object recognition in videos has opened a new door to smarter surveillance systems that can monitor public safety and security. However, the task of identifying moving objects in surveillance videos has its challenges. Accuracy is often affected because objects appear in small sizes with low resolution due to the "surveying" distance of the camera. This also results in perpective distortion of the objects in view. Outdoor surveillance videos which is our focus, offers yet another challenge of unconstrained environmental conditions– lighting, fog, rain, snow. With nature's ever changing ways, objects such as moving tree branches, shadows and illumination changes, camera shakes caused by strong winds are falsely detected. All of which affects the accurate classification of objects from one class to another.

Feature-based approaches is preferred over pixel-based approaches in object classfication due to speed and ease of training. There exist several feature-based approches in literature that we used to compare and combine to ascertain the method with the highest accuracy. In this work, we used a recent large-scale dataset created for long-term outdoor surveillance research called LOST [1] for our object classification.

The aspect of long-term data spanning a few years is a new one for object detection and classification research. At longer time-scales, consistency is hard to maintain, be it natural phenomena like weather, or man-made changes like construction in the captured scene or even the video collection itself where the camera angles slightly with each passing year. These prove to be new challenges that come hand in hand with long-term datasets. Yet tracking and classifying objects in videos under varying weather conditions have received very little attention.

In this paper, we present a object classification framework for long-term video surveillance. Firstly, a suitable approach is introduced for object detection, with additional consideration for the problem of perspective distortion found in wide-area surveillance scenes. Tracks, or trajectories of the extracted moving blobs are then created automatically ensuring that they belong to the similar object. A variety of state-of-the-art feature descriptors are applied individually and in combination to extract meaningful characteristics from the object regions. A standard bag-of-words (BOW) model is used to represent these features. To perform our experiments, two classification schemes, i.e. frame-based and track-based, were considered; the latter purportedly more robust towards environmental variations such as shadows, occlusions or erroneous tracking earlier. We apply the proposed framework on selected videos from the new challenging LOST dataset [1]. On a five-category classification problem, our best performing set of features shows promising results using a standard SVM classifier.

The rest of this paper is organized as follows: Section II reviews some related work in literature. In Section III, we present our classification framework, and elaborate on each stage and their respective choice of methods. Experimental results and discussion are covered in Section IV, while Section V concludes this paper and provides future directions.

II. Related Work

Although object classification (or also known as *object recognition*) has been a long-standing research area for more than a decade, made popular by these early notable works [2], [3], it also found its way into works relating to visual surveillance and monitoring [4]–[6].

An early work by Javed and Shah [4] highlighted a few key problems that often occur in realistic surveillance scenarios which should be dealt with in the detection and tracking steps. Techniques were suggested to perform background subtraction, shadow removal and motion correspondence before proceeding to object classification.

Within the context of surveillance systems, numerous

approaches have been proposed to extract valuable information or features or local patches from a detected/tracked object. Zhang et al. [7] proposed a new feature descriptor called Multi-block Local Binary Pattern (Mb-LBP) to capture large-scale structures in object appearances. In this method all the extracted patches are normalized to a unified scale (20x20 pixels) and converted to a gray-scale patch. AdaBoost learning algorithm is then applied to select the most discriminative features for classification. Chen et al. [8] proposed a method called Differences of Histogram of Oriented Gradients (DHoG) to distinguish a group of people from vehicles by measuring the amount of intra-object deformation frame-by-frame. Another classification work for surveillance [9] employed a two-step training framework– firstly by forming global clusters in the input feature space, and then, applying an online learning algorithm to sequentially learn local clusters from incrementally gathered samples. A local shape spatial pyramid representation, Pyramid of Histograms of Oriented Gradients (PHOG) [10] was used to describe the extracted moving object regions. A 4-class experiment on 18 video sequences (4,000 training samples, 4,240 test samples) reported an impressive result of more than 98%.

With the limitations of unlabeled training data, an unsupervised method [11] was introduced to learn and detect dominant object classes in a generic dynamic scene. Here, HOG [12] features were used to construct the bag-of-features for clustering. Gurwicz et al. [13] presented a set of efficient and robust features for classifying up to five types of objects prominently found in real-world video surveillance scenes: human, body organs (or parts), bag, group of people and clutter. The overall feature set comprise of a combination of five different groups of features: structural features (such as morphological features, luminance asymmetry), smoothed DCT coefficients, central moments, ART moments and Zernike moments. An in-depth selection process was then applied to unify the best groups of features. Experiments that were performed on a relatively large dataset of around 29,000 object instances (out of which 21,000 were for evaluation) chosen from various scenes worldwide yielded a high classification accuracy of 95%.

More recently, Elhoseiny et al. [14] attempted to classify objects detected in outdoor videos from the VIRAT surveillance dataset into five distinct classes (Human, Car, Vehicle, Objects, and Bicycle) using different types of feature extraction techniques, similar to those from [13]. They reported the weaknesses of appearance-based features such as HOG, and how it can become very effective when appropriately combined with geometric features such as luminance symmetry, central moments and ART moments. Due to the large dimensionality of the features used, they resorted to dimensionality reduction to ease the computational burden of their system. Although results were reportedly promising, the authors opted for a seemingly unbalanced training-test split of 80%-20% in order to provide more generalization from samples.

While all these aforementioned works have addressed the problem of recognizing objects in surveillance video to a certain degree of success, their scope does not extend towards the new challenges of long-term surveillance. Though the VIRAT dataset used in [14] is already quite extensive from the aspect of scene and view variability, it does not deal with scenes that are collected over a long period of time, which will present a different challenge altogether. Moreover, VIRAT videos were captured at high-definition resolutions, which is not the case in many real-world scenarios.

The LOST dataset [1] was birthed through an interesting project by Washington University in St. Louis, attempting to collect a large number of outdoor streaming webcams from all over the world, capturing the same half hour each day. Most of its data were collected since 2010 for span of a few years with some cameras still continuing up until today. There are very few works in literature that have worked on surveillance tasks to this degree of duration. A recent work [15] proposed a framework that facilitates the mining of anomalies in the LOST dataset. Meanwhile, another work by Zen et al. [16] proposed a foreground detection technique for analyzing behavioral patterns in complex urban scenes using a month-long of traffic video data (24 hours-per-day) captured from New York City. While the LOST dataset provides useful meta-data, including geo-location, object detection and tracking annotations, much of its object bounding boxes and track trajectories were not accurately detected. As the dataset was originally intended for detection of behavioral anomalies and scene analysis across time, the object and tracks were labeled using a simplistic threshold-based blob detection and linking procedure. On the contrary, the detection of moving objects and track creation have to be accurately performed in our work to enable proper classification of the detected objects.

III. Proposed Framework

In this section, we present the proposed framework for object classification, with in long-term surveillance videos. Although this pipeline may not be uniquely different from those put forth in conventional bag-of-words (BOW) models [17], suitable techniques are devised to accommodate the challenging nature of long-term surveillance scenes. Henceforth, we describe in detail the five main stages in the pipeline shown in Figure 1: Object detection and perspective distortion correction, track creation, feature extraction, feature representation by BOW, and finally, classification.

A. Object Detection & Perspective Distortion Correction

Similar to most related works, the initial step to approach object classification in videos is by segmenting the moving objects out of the scene. We use the background subtraction method in [4], [18] to perform the detection of moving objects. By some morphological operations, spurious noises in the foreground are first eliminated before determining each object's bounding area by grouping the connected pixels.

Due to the wide area of coverage typically encountered in the monitoring of outdoor scenes, perspective distortion is likely to pose a difficult problem to the latter stages of the pipeline. To solve perspective distortion of the extracted blob objects, we compute the corrective coefficient, $c = m * (V_{ref}/V)$ to re-size these blob images. Here, V_{ref} is the middle ordinate of the video frame based on the top-most and bottom-most moving objects in the scene (see Figure III-A), V is the ordinate of a moving object while m is the up-sample factor ($m = 2$ is found to be most suitable). The ordinate of the moving object can be simply obtained by taking the ordinate passing through the middle of the bounding box.

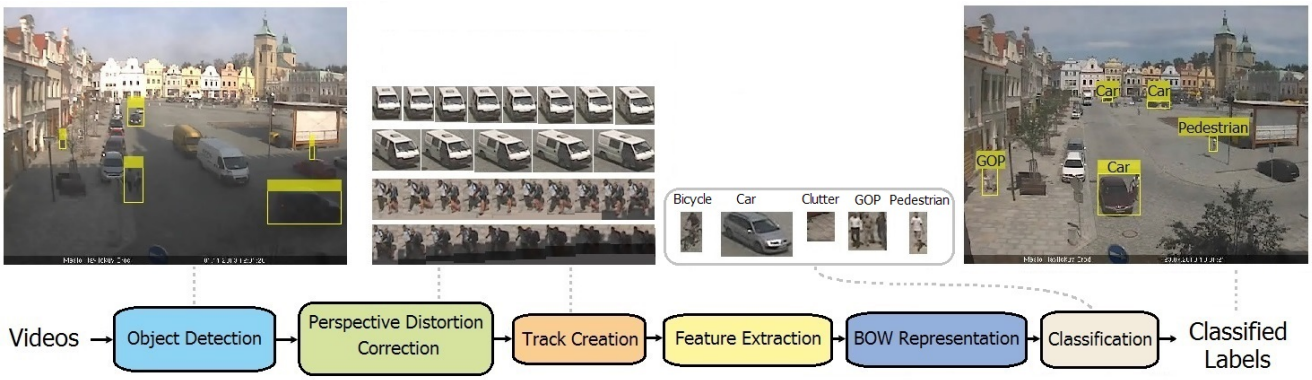

Fig. 1. Proposed framework for object classification in long-term surveillance videos

B. Track Creation

After detecting the moving objects and correcting any perspective distortion, the next step involves piecing together the extracted object regions to form *tracks*. Motion-based tracking using Kalman filter is first applied to predict the object centroids in subsequent frames. To assign relevant frames to a particular track, a method that minimizes the distance cost between all objects from subsequent frames is used. First, it computes the assignment cost as the Euclidean distance between the predicted object centroid of each existing track and the centroid of the detected object. After that, the track with the least total cost can be efficiently computed by the Munkres' Hungarian algorithm. Lastly, tracks that remain unassigned after many consecutive frames are deleted as they might be either too short or potentially a false positive.

C. Feature Extraction

In this work, we performed feature extraction using a variety of state-of-the-art methods, from shape-based descriptors (HOG), to local interest point-based descriptors (SIFT, DSIFT, PHOW) and texture-based descriptor (LBP). Each of these descriptors were implemented individually, while some were combined together by concatenation of feature histogram (see Sections III-D and IV). We now describe each of these descriptors in brief:

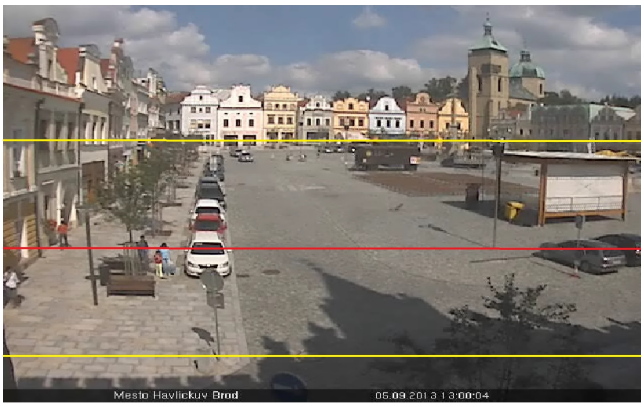

Fig. 2. The ordinates of the top-most and bottom-most moving objects indicated by yellow line. The middle ordinate, which is used as the reference ordinate for correcting object perspective, is indicated by red line.

1) HOG Features: Histogram of Oriented Gradients (HOG) [12] is a gradient-based feature descriptor which is widely used for the purpose of object detection and recognition, specifically for human or pedestrian detection. This technique formulates histograms based on the number of occurrences of gradient values of the localized object. This is able to encode the general shape information of the object.

2) SIFT Features: Lowe [2] introduced scale-invariant feature transform (SIFT), a landmark method which is popularly used to detect and describe local features in an image. Interest keypoints (determined by minimas/maximas of Difference-of-Gaussians) at different scales are extracted from the image and assigned orientations. Histograms are then computed from the magnitude and orientation values around the keypoint from a 4×4 subregion of the original neighborhood region. The SIFT descriptor has a dimensionality of 128.

3) Dense SIFT: The Dense SIFT algorithm [19] provides SIFT features at every location, unlike the normal SIFT algorithm [2], which only provides SIFT features at keypoint locations. Thus, the Dense SIFT descriptor is much larger than the normal SIFT descriptor. The standard SIFT descriptor produces a vector of $128 \times N$ dimension while Dense SIFT produces a vector of $128 \times M$ dimension, where M is almost more than 100 times bigger than N.

4) PHOW Features: Pyramid Histogram Of Visual Words (PHOW), a variant of Dense SIFT extracted at multiple scales in a pyramid structure was introduced by Bosch et al. [20]. In order to form the pyramid, the grid at level l has 2^l cells along each dimension, i.e. level 0 is shown by a N-vector corresponding to the N bins of the histogram, level 1 by a $4N$-vector and etc. As such, the pyramid descriptor of the entire image is a vector with dimension of $N \sum_{l=0}^{L} 4^l$. The original paper suggests for $L = 3$ to prevent overfitting.

5) LBP Features: Local Binary Patterns (LBP) [21] is a feature that encodes texture information in images. The LBP model labels the pixels of each image by thresholding the neighborhood of each pixel by the value of the center pixel. Then, the resulting sequence of binary values are considered together as a binary number. A histogram is then constructed from these labels across the entire image to form the descriptor. Generally, the LBP operator can accommodate circular neighborhoods of different radii, and also different number of sampling points interpolated along the circle. In this work, we apply the standard 3×3

neighborhood (or radius 1) with 8 neighboring pixels along the circular neighborhood.

D. Bag-of-Words Representation

Bag-of-Words (BoW) is a representation model in the field of computer vision made popular by Csurka et al. [22] to represent an image as a *document*, containing a collection of features or *words*. The extracted features are used to generate a codebook (or *vocabulary* in the document analogy) that encodes the most representative features. This is done in a quantization step, typically by performing k-means clustering over the features, resulting in K number of cluster centers representing the codeword (here, K is also referred to as the size of the vocabulary). As such, all features are mapped to their respective codewords, and the image is represented by the histogram of these codewords. Subsequently, features from a new sample can be mapped to the clusters by proximity to the closest centers, in turn producing its histogram.

Following the practice in [23], we hold-out a number of training samples per class, and then randomly sample a maximum of 100,000 features to construct the codebook. Clustering by k-means is repeated for a maximum of 50 times (or less if other constraints are met) to allow better convergence.

E. Classification

For classification, we applied the standard Support Vector Machine (SVM) classifier due to its robustness against outliers [24]. A multi-class SVM is used to build multiple binary classifiers based on a *one-versus-all* strategy and the classifier with the maximum output score is taken as the predicted object class. Due to the computational burden of applying highly effective nonlinear Mercer kernels such as intersection kernel and chi-squared kernel, we employ an efficient explicit feature mapping which was introduced by Veldaldi et al. [25]. These homogenous kernels are able to map to a very good level of approximation to conventional kernels in SVM.

1) Frame-based scheme: Frame-based classification is similar to conventional image classification, whereby the feature vector of each extracted object image $x_i \in \mathbb{R}^d$ of track $T = \{x_1, \ldots, x_{N_T}\}$ is to be identified as belonging to a sample label $y \in 1, \ldots, C$ of a closed set of C object classes. In other words, each extracted object image is considered a test sample for classification.

2) Track-based scheme: For track-based classification, we employ a simple majority voting algorithm to pool together the predicted labels of all the frames in each track, arriving at a single label. Basically, the algorithm counts the number of predictions secured by each class from the track, and takes the most voted class as the final decision. Concretely, the track test sample $T = \{x_1 \ldots, x_{N_T}\}$ is to be assigned the label

$$y_T = \arg\max_j \{|y_j|\} \qquad j = 1, \ldots C \tag{1}$$

where y_j is the corresponding predicted label for frame x_i. We hypothesized that this algorithm is intuitively robust against environmental changes (lighting, shadows, occlusions) that might occur along an object's track by circumventing a small number of misclassifications. For instance in Figure 3, most frames are carrying sufficient information

Fig. 3. Sample track of a group of people (GOP) with severe shadow conditions in some frames on the right-hand-side. Majority voting enables this track to be correctly classified.

for it to be classified as a group of people (GOP), hence the entire track should be labeled as that of GOP. However, if they are treated separately, a portion of the frames will potentially be classified incorrectly. This scheme leverages information collected across time (by the multiple instances of an object) for better class prediction.

Since the LOST dataset did not provide labels for the moving objects present in each of their videos, ground truth annotation is an extremely tedious and difficult process that needs to be done manually. To scope our classification task to a manageable level, we manually label the extracted frames into five distinct classes: Pedestrian, Group of People (GOP), Car, Bicycle, Car.

IV. Experiments and Results

Experiments in this work were conducted on videos from the recently collected LOST dataset [1], possibly the first compilation of outdoor surveillance videos that is long-term in nature. Due to the massive size of the dataset, we selected only a portion of videos from camera '017', which is located at Czech Plaza (Havlickuv Brod, Czech Republic).

A total of 32 videos were selected (from 23 July 2013 until 23 June 2014) for the purpose of object detection and track creation. This one year period provides a large variety of seasonal and weather conditions (sunny, cloudy, rainy, snowy) and also potentially changing behavioral patterns from the moving objects. After the moving objects were detected, corrective prospective distortion is an essential step which is required to remedy the consistency of the size of detected objects before track creation takes place.

The total number of the extracted moving object frames vary between 4,000–9,000 frames for each video; the number of tracks range between approximately 300–2,500

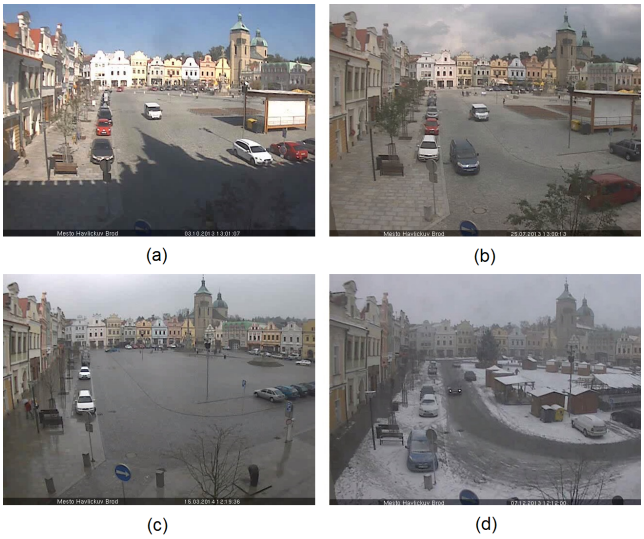

Fig. 4. Sample frames showing different weather conditions for camera '017': (a) sunny, (b) cloudy, (c) rainy and (d) snowy.

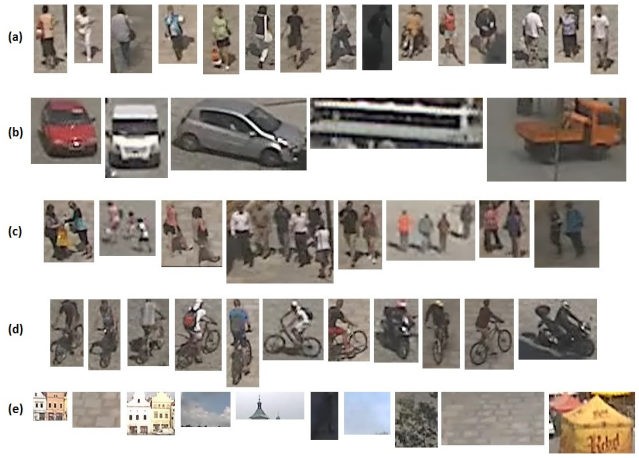

Fig. 5. Sample images of different objects extracted from the LOST dataset videos: (a) Pedestrian, (b) Car, (c) GOP, (d) Bicycle and (e) Clutter.

tracks, depending on other factors within the scene. For example, videos on sunny days typically produce more number of humans and vehicles compared to videos which were captured on stormy or snowy days. However, in scenes with bad weather, most of the extracted moving objects are clutters, consisting of moving clouds, moving shadows of buildings, moving tree branches or other naturalistic changes in the scene. Figure 5 shows sample extracted images of the five different objects extracted from various scenes from camera '017'. Note the high variability of view angles of the human and bicycles, different car types, size of group of people, and the randomness of clutter images. After track creation, good samples from each class (∼1,000 frame instances per class, amounting to ∼5,000 frames instances) were manually chosen.

A. Experiment 1: Frame-based Classification

In our first experiment, we ran a vanilla closed-set[1] evaluation based on frame-based scheme (which considers frames as samples) to find the best feature extraction technique. A few different combination of these features were also tested out. In this simple experiment, we used a total of 1,500 frames (300 frames/class) for both training and testing. For fairness, we ensure that samples from training and test sets do not overlap. Figure 6 compares the frame-based classification results using different feature vectors (individual and concatenated). Overall, the texture-based LBP and shape-based HOG descriptors did not fare well in such complex surveillance scenarios. The best method of PHOW+DSIFT+SIFT achieved a high recognition rate of 89.67%. From its confusion matrix in Table I, we observe that classifying 'Bicycle' samples is the most challenging task in long-term surveillance considering its large variability in appearance and pose.

TABLE I. CONFUSION MATRIX OF FRAME-BASED CLASSIFICATION USING PHOW+DSIFT+SIFT

Classes	Bicycle	Car	Clutter	GOP	Pedestrian
Bicycle	**77.33**	1.0	0.67	10.0	11.0
Car	0	**95.33**	4.33	0	0.33
Clutter	0.67	0.33	**96.67**	0.67	1.67
GOP	3.0	0	1.33	**89.67**	6.0
Pedestrian	3.0	0	1.67	6.0	**89.33**

[1]Only frame samples from the selected 32 videos from camera '017' were used.

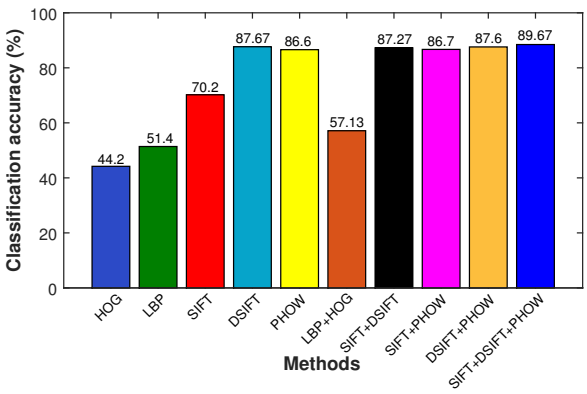

Fig. 6. Performance of different feature methods using frame-based classification. The last 5 bars involved concatenated features.

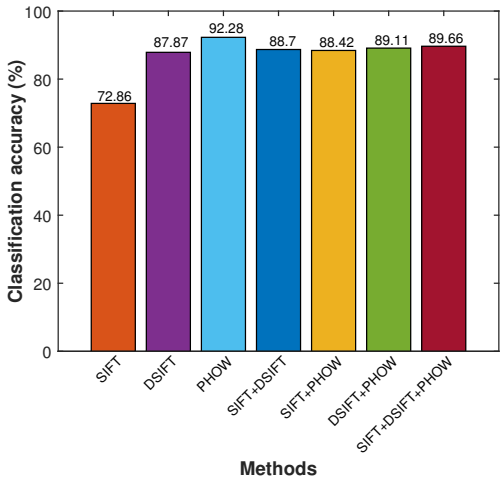

Fig. 7. Performance of different feature methods using track-based classification.

B. Experiment 2: Track-based Classification

In the second experiment, we performed an open-set track-based evaluation, using two new previously unused videos. We select a total of 2,500 frames (500 frames/class) for training, more than that used in Experiment 1 in order to generalize better with the new videos. From the two test videos, 726 tracks were extracted (from 10,164 frames), with the majority of tracks belong to Car and Pedestrian classes, while there are only 11 Bicycle tracks. Figure 7 shows the classification accuracy of different feature vectors (individual and concatenated). Due to the poor performance of the LBP and HOG descriptors in the previous experiment, we decided to leave them out of this experiment.

This round, our best method (PHOW) achieved a good recognition rate of 92.28%, an improvement of nearly 6% over its frame-based classification performance Across the board, most tested methods improve around 1-2% compared to that of the previous experiment. Table II shows improvement in three out of the five classes ('Bicycle', 'Car', 'Pedestrian'). This verifies the robustness of this scheme in handling object variations found in complex, unconstrained scenes.

TABLE II. CONFUSION MATRIX OF TRACK-BASED CLASSIFICATION USING PHOW DESCRIPTOR

Classes	Bicycle	Car	Clutter	GOP	Pedestrian
Bicycle	**90.91**	9.09	0	0	0
Car	0	**97.4**	2.6	0	0
Clutter	0.85	2.54	**93.22**	1.69	1.69
GOP	10.32	0	2.38	**86.51**	0.79
Pedestrian	3.33	0.83	2.92	1.67	**91.25**

TABLE III. TRACK-BASED CLASSIFICATION ACCURACY OF 4 TEST VIDEOS UNDER DIFFERENT WEATHER CONDITIONS

Video Date	Conditions	Frames	Tracks	Accuracy Rate
2013-08-15	Sunny	5124	343	91.54%
2013-09-05	Sunny	5107	383	92.95%
2013-11-28	Rainy	5166	185	81.08%
2014-01-25	Snowy	8398	329	81.76%
Overall		23795	1240	87.82%

C. Experiment 3: Variation in weather conditions

The third experiment shown in Table III reports how our best method (PHOW) fared against different weather conditions. Using tracks extracted from two new videos with rainy and snowy weather, we observed that the accuracy rate drops to around 81%. Future directions point towards the usage of more robust features, or an improved training process that considers tracks from a larger range of weather conditions.

V. CONCLUSION

In this paper, we have proposed an object classification framework for long-term video surveillance, with feasible algorithms and approaches introduced at various stages of the pipeline. A bag-of-words representation consisting of features extracted from the detected objects is used for the classification of objects into five distinct categories. The more sophisticated track-based scheme was found to be more robust than the frame-based scheme in dealing with various environmental variations. Our best performing features demonstrated a promising accuracy of around 92% in an open-set evaluation consisting of 726 tracks from more than 10,000 frames. In future, we intend to conduct a large-scale evaluation across cameras from different scenes/locations, while also attempting to improve classification capability under varying weather conditions.

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